A120604 concise calculus certificate

Bradley Klee, Mech.An.ika Sol (Open AI)

VERIFIED

July 31, 2026

Definition and first terms

$24\,A(x) = 23 + 64\,x + A(x)^8$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(4)*T(x); T=x+7*T^2+56*T^3+280*T^4+896*T^5+1792*T^6+2048*T^7+1024*T^8`.

$a(0), \ldots, a(7) = 1, 4, 28, 616, 15820, 453208, 13894552, 445970128.$

Kernel used throughout: $\rho(u) = u\,(1 - 7\,u^1 - 56\,u^2 - 280\,u^3 - 896\,u^4 - 1792\,u^5 - 2048\,u^6 - 1024\,u^7)$, so $x = \rho(T(x))$.

Typogeometry

`1x4` `{1,1}x28` `{1,{1,1}}x196` `{1,1,1}x224`

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon\ 7,\ \Delta_3\colon\ 56,\ \Delta_4\colon\ 280,\ \Delta_5\colon\ 896,\ \Delta_6\colon\ 1792,\ \Delta_7\colon\ 2048,\ \Delta_8\colon\ 1024.$

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_7) \in \mathbb{Z}^7_{\geq 0} : \sum_{j=1}^{7} j k_j = n - 1\}, \quad K = \sum_{j=1}^{7} k_j,$$

$$a(n) = \frac{4}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_7} 7^{k_1} 56^{k_2} 280^{k_3} 896^{k_4} 1792^{k_5} 2048^{k_6} 1024^{k_7}.$$

$$a(n) = \frac{4}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{7} P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-7 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#/objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 7. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n,u) = N(n,u)/\rho(u)^6,\ \deg_u N = 49.$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n,u) H_n(u))\,.$

Matrix dimensions: 16×16; remainder 7×8, rank 7, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.